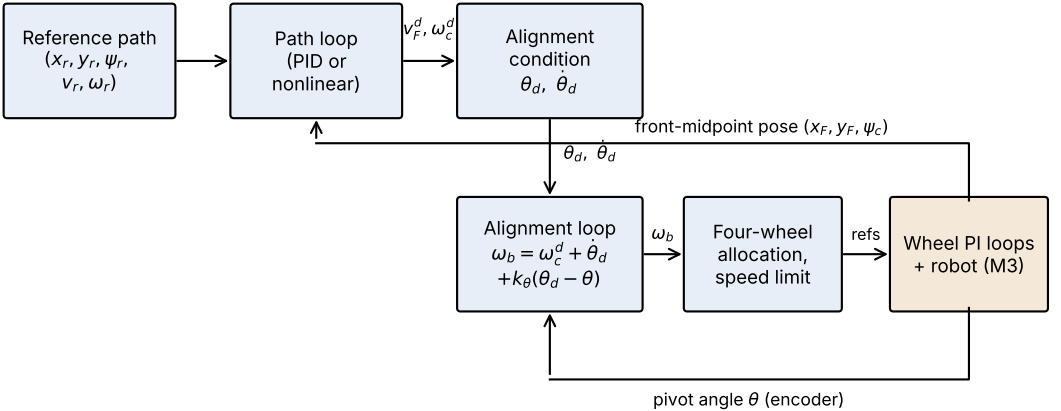
Reference path
$(x_r, y_r, \psi_r,$
$v_r, \omega_r)$
Path loop
(PID or
nonlinear)
v_F^d, ω_c^d
Alignment
condition
$\theta_d,\ \dot{\theta}_d$
front-midpoint pose (x_F, y_F, ψ_c)
$\theta_d,\ \dot{\theta}_d$
Alignment loop
$\omega_b = \omega_c^d + \dot{\theta}_d$
$+k_\theta(\theta_d - \theta)$
ω_b
Four-wheel
allocation,
speed limit
refs
Wheel PI loops
+ robot (M3)
pivot angle θ (encoder)